

Open Compute for Agentic AI (FlagOS)



 [Kernel Operator Development Challenge](#)

24-Hour Speed Kernel Dev

Push the limits of kernel performance optimization and compete for rewards

\$3,000	3	24 Hour	5
Total Prize Pool	Tracks	Timed Challenge	Hardware Platforms

[🔗 3 Tracks](#)

Tracks

Topic

Open Compute for Agentic AI (FlagOS)

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Vice President and Chief Engineer
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Welcome to Open Compute for Agentic AI

Open Compute for Agentic AI · FlagOS

Yonghua Lin

Lead of FlagOS, VP and Chief Engineer of BAAI

Three questions for today:

What kind of compute infrastructure does agentic AI need?

How can open-source communities build it together?

How do we evaluate, govern, and deploy agentic systems in the real world?

Agentic AI creates new compute patterns:

Thinking workload

Long-context reasoning

Multi-step tool calling

Persistent memory and state

Execution runtime

Code execution & sandboxing

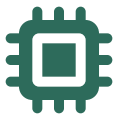
Evaluation in real environments

Always-on inference workloads

Inference becomes the runtime of intelligence.

Important open-weight models released in 2026

Model/Organization	Model information
DeepSeek-V4 / DeepSeek	DeepSeek-V4-Pro (1.6T params, 49B active), DeepSeek-V4-Flash (284B parms, 13B active)
Qwen 3.5 & 3.6 / Alibaba (Qwen)	Qwen3.5-397B-A17B, Qwen3.5-35B-A3B, Qwen3.6-27B, Qwen3.5-Omni
Gemma 4 / Google DeepMind	Gemma 4 (2B, 4B, 26B-A4B, 31B)
MiniCPM / ModelBest	MiniCPM-o 4.5 (9B)
Kimi K2.5 & K2.6 / Moonshot AI	Kimi-K2.5 (terascale MoE), Kimi-K2.6
GLM-5.1 / Z.ai	GLM-5.1
Nemotron 3 / NVIDIA	Nemotron 3 Super, Omni, VoiceChat
Granite 4.1 / IBM	Granite 4.1 (3B, 8B, 30B)
Xiaomi MiMo V2.5 / Xiaomi	MiMo-V2.5, MiMo-V2.5-Pro
Sarvam 30B / 105B / Sarvam AI	Sarvam-30B, Sarvam-105B
SenseNova U1 / SenseTime	SenseNova-U1
Mistral Medium 3.5 / Mistral AI	Mistral-Medium-3.5-128B
Hunyuan HY-3.0 / Tencent	HY-3.0-preview (295B total, 21B active)
Phi-4 / Microsoft	Phi-4-Reasoning-Vision-15B
MiniMax M2.5 & M2.7 / MiniMax	MiniMax M2.5 (Agent-focused), MiniMax M2.7 (Self-evolving)



More chips

AI systems can run on more different available chipsets



More choices

Developers can choose the right hardware for the right workload



Lower costs

Agentic AI can move from demos to real-world deployment

Open compute turns open-weight models into real deployment choices.

Control · Cost · Privacy · Resilience · Innovation

Let's use DeepSeek V4 as example

DeepSeek V4 shows why frontier open models need open compute.

Why DS-V4 matters for

Agentic AI

- 1M-token context
- MoE architecture
- Tool calling and interleaved thinking



What DS-V4 requires

- Hybrid CSA/HCA attention
- Heterogeneous KV cache
- FP4 / FP8 precision
- New kernels and runtime optimization



Why deployment becomes hard?

- Only chips top-end AI chips supports FP4/FP8 precision.
- Requiring >80GB memory for single device.
- Kernels remain coupled with specific hardware backends.

**DeepSeek V4 shows why open compute matters:
frontier open models need a portable path to deployment across chips.**

 What is FlagOS? <https://github.com/flagos-ai>

FlagOS 2.0 is a unified, open-source AI system software stack that enables AI applications to run seamlessly across diverse hardware platforms.

18
Chip Vendors

32
AI Chips

Various AI models

Language models Multimodal models World models Embodied AI

FlagOS Ecosystem-enabled modules

Megatron-FL

TransformerEngine-FL

vLLM-plugin-FL

VeRL-plugin-FL

SGLang-plugin-FL

Pytorch-plugin-FL

Domain solution

FlagOS-Robo

FlagOS Skills

Deploy, code-gen, migration, benchmark

FlagOS core libraries

FlagGEMs operator libraries

FlagGEMs, FlagBLAS, FlagSparse, FlagFFT, FlagTensor, FlagAudio

FlagTree Unified AI compiler

FlagScale parallel framework

FlagCX Unified comm. lib

FlagOS tools

KernelGen: operator auto-gen tool

FlagRelease: Model auto-migration tool

FlagPerf: AI chip evaluation tool

Servers of different AI chipsets



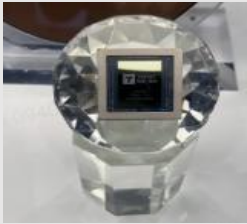
Edge devices



AI book



ARM devices RISC-V AI chips



Key technologies to enable DeepSeek V4 onto various AI

- **FlagGems:** full operator replacement
- **KernelGen:** rapid generation of new operators
- **FlagTree / Triton-TLE:** write once, compile across chips
- **vLLM-plugin-FL:** multi-chip inference integration
- **FlagCX:** o-group parallelism, communication & parallel strategy
- **FlagRelease:** release once, run across chips

Have enabled DeepSeek v4, Minimax M2.7, Qwen3.6, Minicpm-4-o on various AI chips

FlagGems: the Largest Multi-Chip Triton Operator Library, 500+ Operators

FlagGems: 430+ AI operators

Largest single Triton operator library

- FlagGems is a PyTorch Foundation ecosystem proj.



- 18 important vLLM fused operators supported

Inference operator coverage

90%–100% coverage across 40 mainstream model inference tasks

C++ Wrapper adopted by multiple chip vendors to reduce Triton invocation overhead

6 new Triton domain libraries: 90 operators

FlagOS expands from large model-specific operators to broader scientific computing coverage.

FlagDNN

Deep neural network

10

FlagBLAS

Basic LinAlg

30

FlagFFT

Fast Fourier

2

FlagSparse

Sparse

22

FlagTensor

Tensor

FlagAudio

Speech

Full Operator Replacement for DeepSeek V4

DeepSeek V4 originally uses ~67 CUDA + TileLang operators

- FlagGems covers all 67 operators
- MoE scheduling, Attention, RMSNorm and TopK are implemented with Triton / Triton-TLE
- C++ Wrapper reduces Triton launch overhead
- Supports 10 classes of chip backends for DeepSeek V4

Supports 16 vendors 28 chips include NVIDIA, Huawei Ascend, Mthread, MetaX etc.

<https://github.com/flagos-ai>

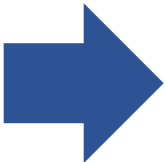
From N×M Adaptation to N+M Collaboration:

The problem:

N chips · M frameworks · Thousands of models · Fast model iteration

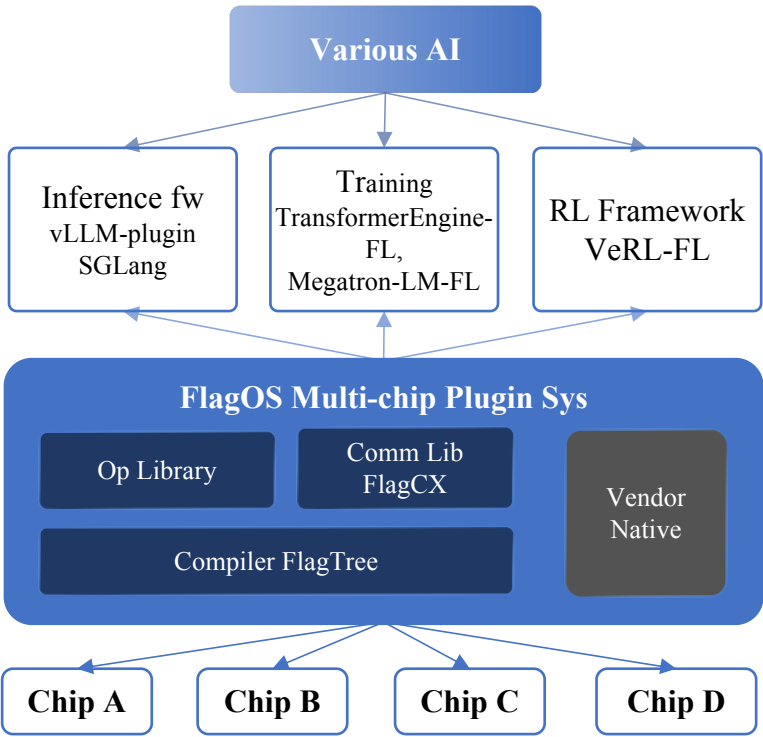
The old way:

Every vendor patches every framework



The open compute way:

Unified plugins turn N×M into N+M



FlagOS plugin system:

vLLM-plugin-FL

<https://github.com/flagos-ai/vllm-plugin-FL>

SGLang-plugin-FL

Pytorch-plugin-FL

Megatron-LM-FL

<https://github.com/flagos-ai/Megatron-LM-FL>

TransformerEngine-FL

<https://github.com/flagos-ai/TransformerEngine-FL>

VeRL-FL

Collaboration with “M” framework communities are very important for Open Compute

FlagOS Day0 Migrated DeepSeek V4 onto 10 different AI Chips

FlagOS enables DeepSeek V4 on 10 different AI chipsets, including hardware without FP4/FP8 and with ≤64GB accelerator memory.

Mode	Chip/Platform	Precision
DeepSeek-V4-Flash	NVIDIA (H20/H100)	BF16 / FP8
	MThread	FP8
	Hygon/Metax/Huawei/PPU/Kunlun/iluvatar/SunRise/TsingMicro	BF16
DeepSeek-V4-Pro	NVIDIA (H20/H100)	Int8 / FP8
	MThread	FP8
	Hygon/Metax/Huawei	Int8

One-stop code for DeepSeek-V4 with FlagOS
<https://github.com/flagos-ai/DeepSeek-V4-FlagOS>

Inference code and with tools

- O-Groups Grouped Projection Communication
- FP8/FP4 → BF16 Weight Conversion Tool
- INT8 MoE Expert Quantization
- Streaming Weight Conversion (Memory-Optimized)

One-stop model files for 10 AI chips:
<https://huggingface.co/flagrelease>

< FlagRelease's models 121 🔍

⬆️ Sort: Recently updated

FlagRelease/DeepSeek-V4-Pro-nvidia-FlagOS
Updated 34 minutes ago

FlagRelease/DeepSeek-V4-Pro-hygon-FlagOS
Updated about 1 hour ago

FlagRelease/DeepSeek-V4-Flash-tsingmicro-...
Updated about 14 hours ago

FlagRelease/DeepSeek-V4-Pro-mthreads-Flag...
Updated 1 day ago

FlagRelease/DeepSeek-V4-Pro-metax-FlagOS
Updated 2 days ago

FlagRelease/DeepSeek-V4-Pro-ascend-FlagOS
Updated 2 days ago

FlagRelease/gemma-3-1b-it-plugin-FlagOS
1.0B · Updated 3 days ago · 12

FlagRelease/gemma-3-1b-it-FlagOS
1.0B · Updated 3 days ago · 33

FlagRelease/DeepSeek-V4-Flash-iluvatar-Fl...
Updated 3 days ago

FlagRelease/DeepSeek-V4-Flash-kunlunxin-F...
Updated 3 days ago

FlagRelease/DeepSeek-V4-Flash-zhenwu-Flag...
Updated 3 days ago

FlagRelease/DeepSeek-V4-Flash-hygon-FlagOS
Updated 3 days ago

FlagRelease/DeepSeek-V4-Flash-metax-FlagOS
Updated 3 days ago · 1

FlagRelease/DeepSeek-V4-Flash-ascend-Flag...
Updated 3 days ago · 1

FlagRelease/DeepSeek-V4-Flash-mthreads-Fl...
Updated 3 days ago

FlagRelease/DeepSeek-V4-Flash-nvidia-Flag...
Updated 3 days ago · 1

With FlagOS, we have also supported

Qwen3.6 on 7 AI chips · MiniCPM-o 4.5 on 6 AI chips

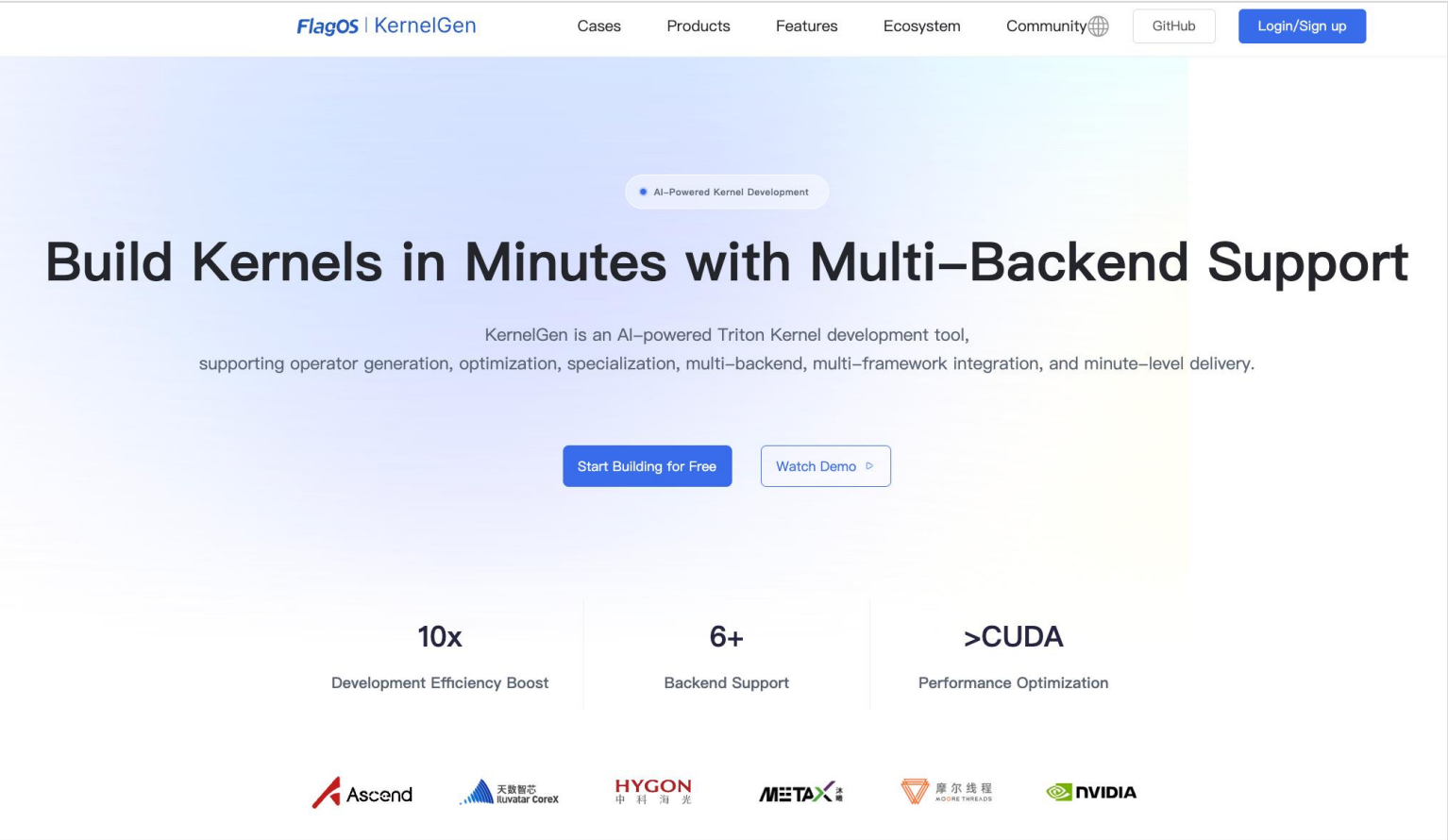
MiniMax 2.7 on 6 AI chips · DeepSeek V4 on 10 AI chips

in last 3 months

KernelGen is an automatic operator generation system for new models on vairous AI chips, covering the complete lifecycle including **code generation, test case generation, precision/speed-up testing, continue optimization.**

Supported Languages & Chips

- Languages: Triton, Triton-TLE
- Chips: Huawei Ascend, MThreads, Hygon, iluvatar, MetaX, NVIDIA
- Interfaces: VS Code, Claude Code, OpenClaw
- Products: Web platform, Agent Skills, MCP



Performance Highlights

- **NVIDIA:** Significantly outperforms Claude Code in correctness and speedup (110 operators on KernelGen-bench)
- **The other 5 AI chips:** >92% operator generation capability, >50% operators exceed its native performance in C language.

KernelGen 2.0 vs Claude Code (LayerNorm Example)

Metric	KernelGen 2.0	Claude Code
Execution Time	14m10s ↓	34m42s
Token Consumption	10K ↓	21.6K
Accuracy	100%	100%
Speedup	1.14x ↑	0.64x

10+ Skills covering model adaptation, operator generation, performance tuning, and other scenarios for domestic AI chips

Deployment & Release

- model-migrate-flagos
- install-stack-flagos
- gpu-container-setup-flagos
- flagrelease-entrance-flagos

Benchmarking

- model-verify-flagos
- perf-test-flagos

Operator Development

- tle-developer-flagos
- kernelgen-flagos
- Kernelgen-for-flaggems
- kernelgen-for-vllm

Skill Example

Helps users automatically complete new model adaptation to FlagOS vllm-plugin, with 13 steps in total, ultimately completing accuracy comparison with the original model to obtain multi-chip support.

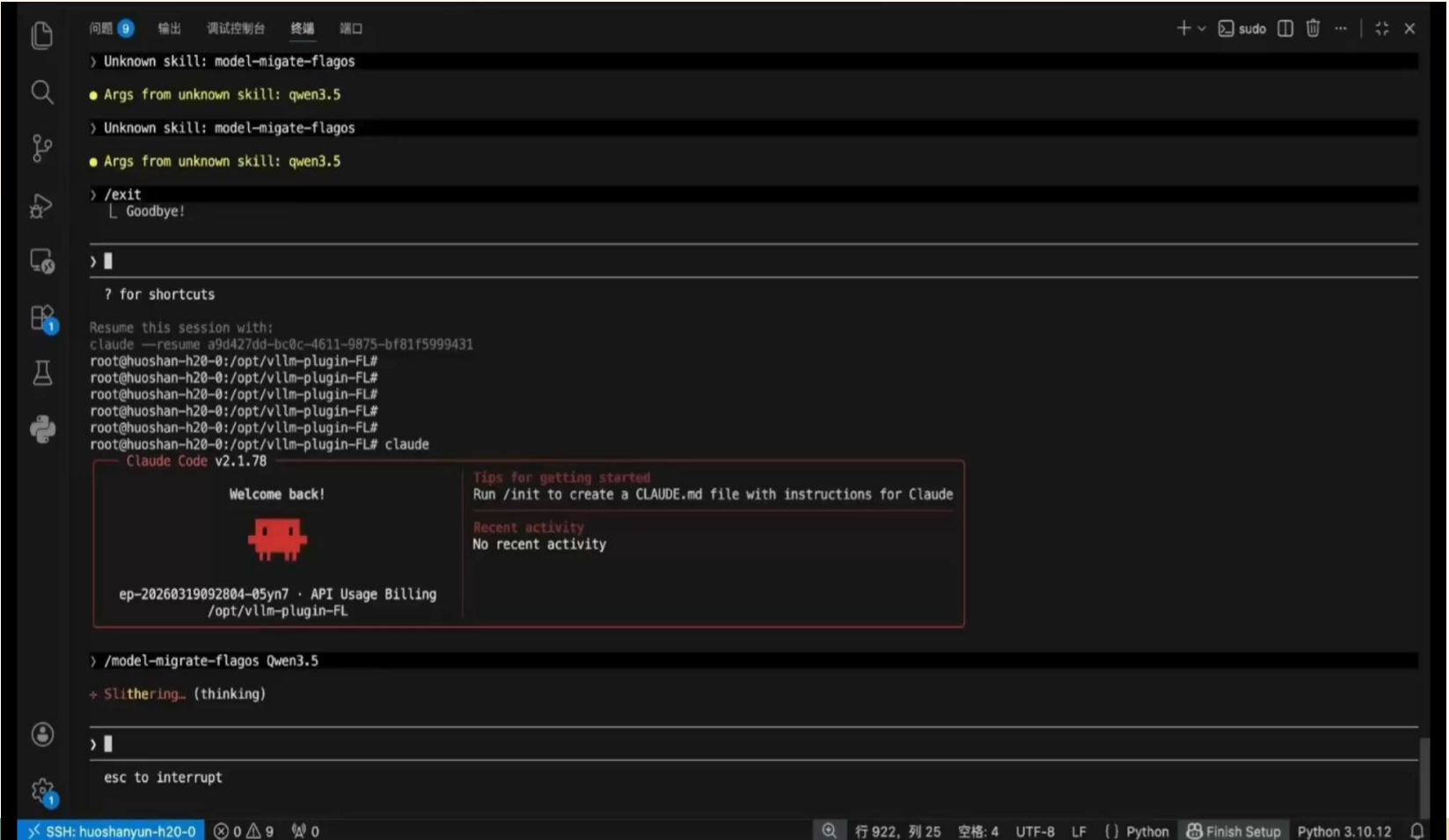
GitHub: <https://github.com/flagos-ai/skills>

model-migrate-flagos: vLLM Model Migration Skill

[中文版](#)

Overview

`model-migrate-flagos` is an AI coding skill that migrates model code from the latest vLLM upstream into the `vllm-plugin-FL` project (pinned at vLLM v0.13.0).



For agentic AI, "it runs" is not enough. We must ask:

- ? Does it produce **trustworthy** results?
- ? Does it follow **rules and constraints**?
- ? Can it be **evaluated in real environments**?
- ? Can it serve **public and high-trust domains**?
- ? Can we **govern it responsibly**?



Open compute must be connected with open evaluation and open governance.

Today's agenda includes **PanEval** open-source evaluation, **AI compliance and governance**, and place-based evaluation testbeds for high-trust public domains.

Open-source Contributors: > 600

Weekly downloads: >10K

Community organization members: >80

Supported AI chipsets: >30

Open source community and Standardization:



ITU standard for unified communication library (FlagCX) across AI chips



FlagGems became PyTorch ecosystem project



PanEval collaboration for open models

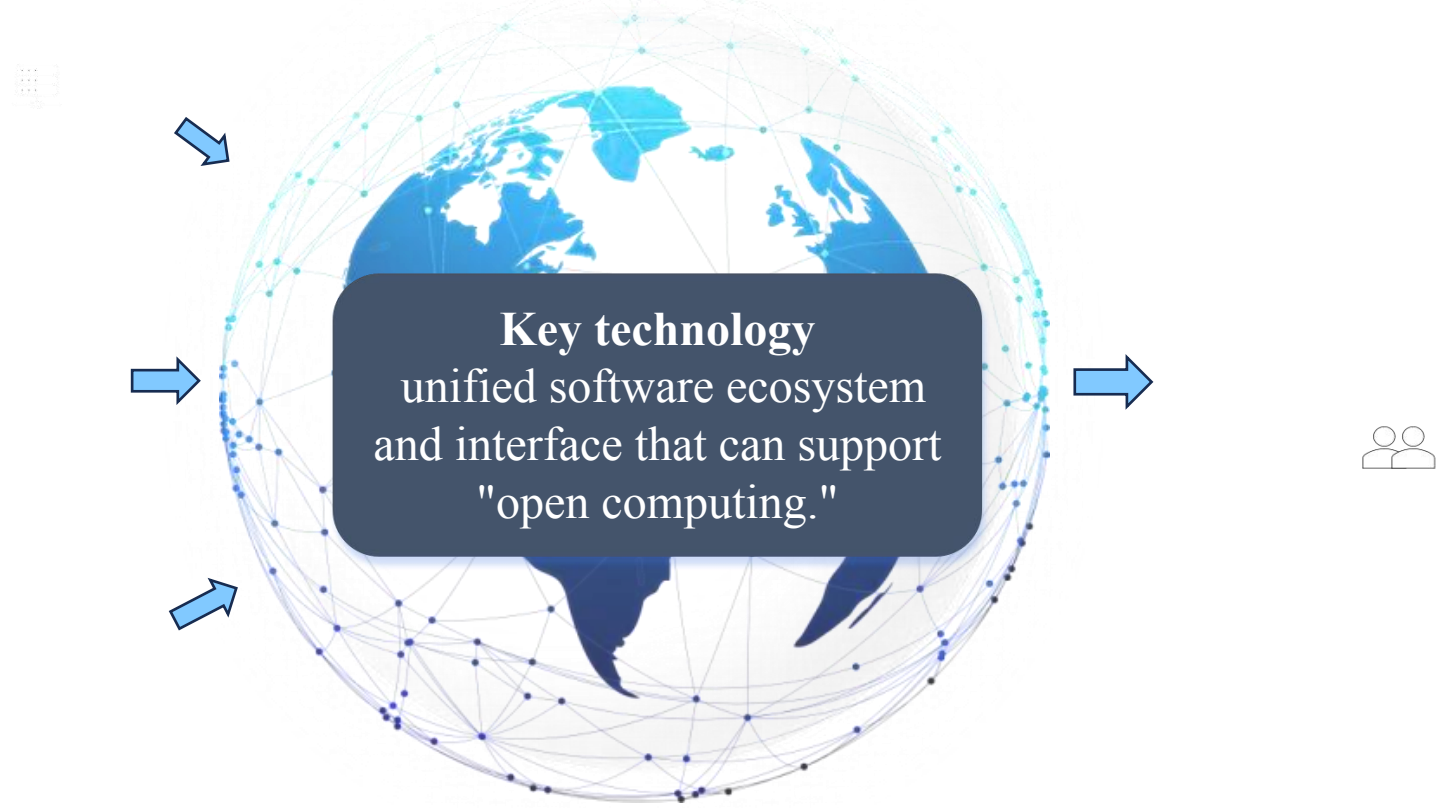
Global Open Compute Initiative for AI : We are looking for broader collaboration



In last one year, we bring it together, and make it happening

- UN Office for Digital and Emerging Technologies
- FlagOS community 
- Beijing Academy of Artificial Intelligence 
- Peking University 
- Africa Union – Africa Science and Research Innovation Council 
- UNESCO 

Global Open Compute Initiative for AI: Open Compute turns underutilized AI compute resources worldwide into accessible infrastructure for developing regions, enabling more inclusive global AI innovation.



The first program: China-Africa AI Talent Development Program

BAAI, AU-ASRIC and Peking Univ. together launched the program and invited 11 teachers from different universities from 10 Africa countries. (Apr.15~Apr.30, 2026)



“Open Compute – online lab” powered by FlagOS (>100 AI devices from 3 vendors)



[Home](#) [Software Stack](#) [Tools](#) [Docs](#) [Learn](#) [Event & Competition](#) [Online Laboratory](#) [User](#)

FlagOS Community offers a series of AI courses and welcomes universities/institutions to apply for collaboration with experimental computing support.

[Submit Collaboration Application](#)

The environment is a containerized environment. Once released, all data will be destroyed and cannot be recovered. Please make sure to back up your data in advance.

Welcome, User [User Guide](#)

DevEnvironment Name	Status	Computing Resource Information	Create Time	Release Time	DevEnvironment Access
activity_29jvhzy9_8...	Running	GPU:Ascend 910C x 2 Card(s) CPU:20 Core(s) System Storage:200G Memory:80G	2026-04-15 15:55:24 (UTC+8)	2026-04-16 00:00:00 (UTC+8) Countdown: 02 hour 53 minute	Online Laboratory Cloud DevEnvironment Secret Key: zSLJz... Management Enter IDE

GOSIM Paris 2026

Open Compute for Agentic AI (FlagOS)

Central Room

May 5

Agenda

14:00 - 14:15

Open Compute for Agentic AI (FlagOS)

Yonghua Lin Vice President and Chief Engineer
Beijing Academy of Artificial Intelligence (BAAI)

14:15 - 14:45

PanEval Open-Source Introduction

Qigang Zhu Head of Open Source Engineering
Beijing Academy of Artificial Intelligence (BAAI)

14:45 - 15:15

AI Compliance and Governance

Michael Plagge Vice President, Eclipse Foundation

15:15 - 15:35

Bridging AI Evaluation and Real-World Deployment:
Place-Based Evaluation Testbeds for High-Trust
Public Domains

Dr. Gábor Soós Hungarian National Commission for UNESCO

15:35 - 15:55

KernelGen 2.0
Auto Operator Generation & Benchmark Co-development

Jian Tao Algorithm Engineer, AI Systems Research Group
Beijing Academy of Artificial Intelligence (BAAI)

15:55 - 16:15

Introduction of LF AI & Data Foundation and Omni-AI
community

Zhipeng Huang Board Chairperson, LF AI & Data Foundation
Huawei

16:15 - 16:35

SGLang Project and Roadmap

Xinyuan Tong SGLang Developer

16:35 - 17:00

Open Discussion & Ecosystem Roundtable
(Call for Participation)

Open Call

17:00 - 17:30

FlagOS KernelGen Virtual Coding Competition &
Live Demo

Jian Tao Algorithm Engineer, AI Systems Research Group
Beijing Academy of Artificial Intelligence (BAAI)

Let’s explore together, and build it together

- What kind of compute infrastructure does agentic AI need?
- How can open-source communities build it together?
- How do we evaluate, govern, and deploy agentic systems in the real world?

FlagOS

Open Software for AI Systems

KernelGen

Bounty

Challenge

24 Hour

Build high-performance AI
kernels. Compete.
Win rewards.

A 24-hour operator optimization challenge powered by KernelGen
for heterogeneous AI hardware (GPU / AI accelerators).

3 Operators

3 Leaderboards

Real-Time Ranking

Cash Prizes

Submission Phase

start:

May 5, 2026 - 16:00

(Paris Time)

Deadline:

May 6, 2026 - 16:00

(Paris Time)

Kickoff Talk

May 5, 2026

17:00 - 17:30

(Paris Time)

GOSIM Paris

Awards Ceremony

May 6, 2026 - 17:00

(Paris Time)

On-site at
GOSIM Paris



An Invitation: Let's Build Open Compute Together

GitHub: <https://github.com/flagos-ai>
<https://flagos.io>

Join the discussion · Join the challenge · Bring your work

Open models need open compute.

Open agents need open infrastructure.

Open infrastructure needs open collaboration.